

Late-time growth of an inhomogeneous, turbulent mixing layer subjected to transient compression

Bradley Pascoe ^{a,b,*}, Michael Groom ^{b,c}, Ben Thornber ^{b,d}

^a Institute for Advanced Engineering and Space Science, University of Southern Queensland, Toowoomba, 4035, QLD, Australia

^b School of Aerospace, Mechanical and Mechatronic Engineering, University of Sydney, Sydney, 2006, NSW, Australia

^c Geophysical Fluids Team, CSIRO Environment, Eveleigh, 2015, NSW, Australia

^d Department School of Mechanical and Aerospace Engineering, Queen's University Belfast, Belfast, BT9 5AH, Northern Ireland

ARTICLE INFO

Editor: Dmitry Pelinovsky

Keywords:

Turbulent mixing
Richtmyer-Meshkov
Large-eddy simulation
Multispecies
Strain

ABSTRACT

Recent engineering models of turbulent mixing layers under strain imply that there may be a permanent modification of important mixing layer physics following a temporary application of strain. This paper presents a set of Implicit Large-Eddy Simulations of a canonical Richtmyer-Meshkov instability to explore the validity of this model result. The well-characterised θ -group quarter scale case is modified to include a strain rate which halves domain widths in approximately five eddy turnover times. Following the removal of strain, the observed value of the growth rate exponent $\theta = 0.112$ which is a 2.5 times reduction compared with the unstrained case. Whilst θ is slowly rising at late time, the actual change in θ is qualitatively in good agreement with the engineering model but is quantitatively a much greater change than expected. Mixedness also increases significantly, from $\Theta = 0.8$ for the unstrained to $\Theta = 0.9$ following the application of strain. Turbulent kinetic energy substantially rises during strain, but then dissipates more rapidly following the removal of strain due to the decreased turbulent length-scales. Overall these results demonstrate that modifications to engineering models, such as those proposed by Pascoe et al. (Phys. Rev. Fluids 10 (6), 064609, 2025) are needed to capture these significant variations in flow physics which persist even following the removal of strain. The engineering model further predicts more substantial impacts at high overall compression or expansion as expected in typical applications in inertial confinement fusion, supernovae or explosions.

1. Introduction

Hydrodynamic instabilities such as the Richtmyer-Meshkov instability (RMI) and Rayleigh-Taylor instability (RTI) play a critical role in high-energy-density flows [1–4]. These instabilities arise at accelerated density interfaces and are widely encountered in inertial confinement fusion (ICF) implosions [5]. For example, in an ICF capsule a series of converging shock waves compress the fuel pellet, amplifying interface perturbations via RMI/RTI growth and seeding turbulent mixing that can quench fusion yield. Understanding how compressive strain (e.g. from convergence during the implosion) modifies the growth of RMI-induced mixing layers is therefore of great scientific and practical importance in these applications.

The bulk compression of the mixing layer depends upon the divergence of the velocity field, which is the sum of the normal strain rates of the flow. This is evident from the equation for the total derivative of density,

$$\frac{D\rho}{Dt} = -\nabla \cdot \mathbf{u} = -tr(\mathbf{S}), \quad (1)$$

where $\mathbf{u}$ is the velocity vector, and $\mathbf{S} = \frac{1}{2}(\nabla \mathbf{u} + (\nabla \mathbf{u})^T)$ is the strain rate tensor.

RMI and RTI are anisotropic and exhibit different behaviours depending upon the direction. Frequently, the mixing layers will have a degree of symmetry, showing isotropic behaviour in a given plane which corresponds to the alignment of the initial interface between the two fluids. This is labelled the transverse or spanwise direction for planar configurations, or circumferential in spherical geometry. The direction of the mixing layer growth is referred to as the streamwise or axial direction for planar, and radial for spherical flows.

The mixing layer can experience strain rates either from wave transmission, geometry deformation, or from converging/diverging processes. Analysis of implosion and explosion profiles show that during a

* Corresponding author.

E-mail addresses: bradley.pascoe@sydney.edu.au (B. Pascoe), b.thornber@qub.ac.uk (B. Thornber).

bulk compression or expansion process, the strain rates experienced by the mixing layer are not isotropic between the transverse and axial directions, and can have opposite signs during a shock-driven implosion process [6,7]. The radial/axial compression of the mixing layer for a shock-driven implosion primarily occurs during the shock transmission, otherwise the fluid tends to expand radially. However, the mixing layer is moving inwards throughout and experiences circumferential/transverse compression. The transverse strain experienced by the mixing layer is therefore not proportional to the axial strain experienced, justifying the analysis of the independent effects of transverse strain.

Recent efforts have sought to isolate and understand the influence of imposed strain rates on RMI mixing layers in controlled settings. In practical applications of converging/diverging hydrodynamics, the mixing layers can experience other effects such as re-shock, buoyancy production, non-ideal equations of state, reactions, and radiation. Modelling efforts for such systems can be improved by better understanding the individual contributions of each, including the role of the strain rates during the compression/expansion process. The computational setup utilised here removes many of the complexities of converging or diverging flow - allowing in particular a very controlled application of strain rates and a careful tailoring of the mesh to appropriately resolve the flow through a given strain profile.

Axial strain (i.e. a bulk velocity gradient normal to the interface) was studied by Pascoe et al. [8] in planar RMI simulations. In that work, a uniform compression or expansion was applied along the streamwise direction to mimic the effect of convergent implosion or divergent explosion flows without the geometrical complications that are imposed when simulating convergent geometries. Simulations of RMI-induced turbulence under axial strain revealed that the influence of axial strain that is observed in the linear regime (i.e. modifying the growth rate due to the added background velocity) is partially offset by altered turbulence production: the imposed compression or expansion generates shear that changes the bulk entrainment and turbulent kinetic energy (TKE) distribution. Consequently, the mixing layer's growth under axial compression/expansion is less reduced/enhanced than linear theory suggests. The strained cases also did not attain the same self-similar state as the unstrained case, exhibiting differences in mixedness and anisotropy. Pascoe et al. also showed that a standard buoyancy-drag model (calibrated to unstrained growth) could not fully capture these effects; by introducing new terms to account for strain-induced shear production, the model's accuracy was improved.

In a follow-up study, Pascoe et al. [9] investigated transverse strain effects by contracting or expanding the domain perpendicular to the flow (i.e. imposing a time-varying cross-sectional area). This scenario is analogous to compression or expansion in the polar or azimuthal direction in spherical geometry. Interestingly, the transverse strain was found to have an opposite mixing layer growth trend in the linear versus nonlinear regime: in the linear regime, a planar transverse compression amplifies RMI growth (in agreement with linear theory), but in fully turbulent layers the same compression reduces the growth rate. Under transverse compression, the mixing layer grows at a significantly lower rate than the unstrained case and becomes noticeably more mixed (homogenised). The added transverse velocity gradients increase shear production in the spanwise directions, causing the turbulent kinetic energy to redistribute and deviate from the usual anisotropic state. By adjusting the Youngs & Thornber [10] buoyancy-drag model to include a strain-dependent drag length scale, Pascoe et al. were able to improve its predictions of the evolution of the layer width under transverse strain.

The aforementioned studies are not the only examples that propose buoyancy-drag models, used to predict the mixing layer width W and velocity $V = dW/dt$, which account for the change in the wavelength. In the analysis of supernova structure in spherical geometry [11], the velocity evolves according to,

$$\dot{V} = \mathcal{A}g(t) - C_D \frac{V^2}{\lambda(t)}. \quad (2)$$

This model includes a buoyancy term which depends upon the Atwood number $\mathcal{A}$ and the interface acceleration $g(t)$. The usage of the time-varying wavelength $\lambda(t)$ will modify the solution when the spherical interface converges or diverges. For a fixed wavelength, either pre- or post-strain application, the solution for the mixing layer width, ignoring the radial/axial strain rates, will not reproduce the power-law growth rate and is instead,

$$W = \frac{\lambda}{C_D} \log(\lambda(t)c_1 + C_D t) + c_2 \quad (3)$$

for some coefficients c_1 and c_2 which depend upon the initial conditions. The buoyancy-drag model for spherical implosions in Ref. [12] evolves the velocity outside of the linear regime according to,

$$\dot{V} = -\frac{1-\theta}{\theta} \frac{V^2}{W - d\bar{\lambda}(t)} \quad (4)$$

where d is an additional coefficient, and θ is prescribed by the model. The main purpose of the coefficient d is to modify the drag term to better transition from the early time behaviour (which uses a different expression). Again, the growth is modified while the wavelength is changing, but for a fixed $\bar{\lambda}$ the width grows self-similarly and is given by,

$$W = d\bar{\lambda} + (c_1 + c_2 t)^\theta. \quad (5)$$

In [9], a different drag term was found to perform better for cases under the application of transverse strain, instead using

$$\dot{V} = -\frac{1-\theta}{\theta} \frac{V^2}{W\Lambda_T - d\bar{\lambda}(t)}. \quad (6)$$

With the dependence upon the transverse expansion factor Λ_T , the self-similar growth is modified,

$$W = d\bar{\lambda}_0 + (c_1 + c_2 t)^{\hat{\theta}}, \quad (7)$$

where the modified growth rate $\hat{\theta}$ is

$$\hat{\theta} = \theta \frac{\Lambda_T}{1 + \theta(\Lambda_T - 1)}. \quad (8)$$

The effect of the expansion factor causes the growth exponent to increase with expansion and decrease with compression, as observed for the results conducted. Of substantial significance, this model does not predict the modified growth rate will change if the layer no longer experiences strain, it will remain a fixed function of the original θ and the overall compression during the applied strain. The modified growth rate $\hat{\theta}$ in Eq. (8) is a function of θ and Λ_T . Taking the Taylor series of this function about $\Lambda_T = 1$ gives the approximation,

$$\hat{\theta} = \theta + \theta(1-\theta)(\Lambda_T - 1) + \mathcal{O}(\Lambda_T - 1)^2. \quad (9)$$

As θ is sub-linear, the first order term shows that the resulting $\hat{\theta}$ is positively correlated to the change in Λ_T . In Fig. 1, the values of $\hat{\theta}$ are plotted for several different initial θ values as a function of Λ_T . At $\Lambda_T \ll 1$, the growth rate is diminished, asymptotically approaching zero. For ICF cases which can converge by factors of 12 to 40 [13,14], this could greatly diminish the growth rate associated with RMI. However, in these cases there also exists the strain rate in the direction of mixing which is not accounted for in this model. The results of this simple model may also find application in shock-tube experiments exploring mixing layer development, where there may be small but non-negligible strain rates such as those due to boundary layers. For example, Brouillette and Bonazza [15] highlight that boundary layers may account for between 5–15% of the width of a shock tube by the end of an experiment.

This motivates the present study, which examines the late-time growth of an RMI-induced turbulent mixing layer under a period of imposed compressive strain and its subsequent removal. The remainder of this paper is structured as follows. In Section 2, the details of FLAMENCO, the code used to conduct the simulations, is outlined alongside the specification of the simulation initial conditions and strain application. In Section 3, the simulation results are presented, comparing the development of the unstrained and strained turbulent mixing layers after the application of strain. These results and conclusions are summarised in Section 4.

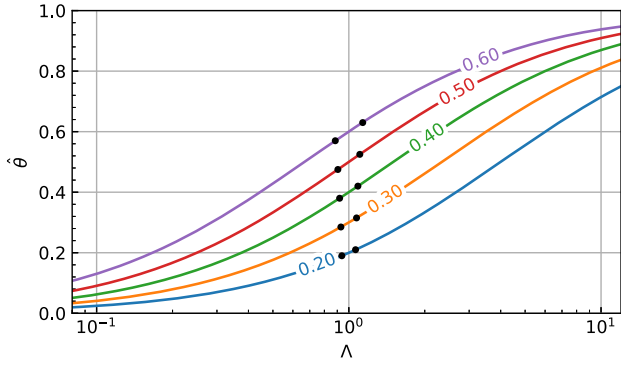


Fig. 1. Variation of RMI growth rate θ as a function of the expansion factor Λ . Black dots indicate the points at which the value of θ has varied by 5% from the unstrained case.

2. Computational setup

2.1. Governing equations

The implicit large eddy simulations are conducted using the inviscid, five-equation, quasi-conservative model of Allaire et al. [16]. In addition to the transport equations for density ρ , velocities $\mathbf{u}$, and total energy E , the model includes transport equations for the mass fractions Y_a and the volume fraction f_a for species a . In the absence of viscous, diffusive, or conduction terms, the equations describe the high-Reynolds number limit of the flow, with unity Schmidt and Prandtl numbers.

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0 \quad (10)$$

$$\frac{\partial \rho \mathbf{u}}{\partial t} + \nabla \cdot (\rho \mathbf{u} \mathbf{u}^T + p \mathbf{I}) = \rho \mathbf{g} \quad (11)$$

$$\frac{\partial \rho E}{\partial t} + \nabla \cdot ((\rho E + p) \mathbf{u}) = \rho \mathbf{g} \cdot \mathbf{u} \quad (12)$$

$$\frac{\partial \rho Y_a}{\partial t} + \nabla \cdot (\rho Y_a \mathbf{u}) = 0 \quad (13)$$

$$\frac{\partial f_a}{\partial t} + \mathbf{u} \cdot \nabla f_a = 0 \quad (14)$$

The total energy E consists of the internal energy e and kinetic energy,

$$E = e + \frac{1}{2} \mathbf{u} \cdot \mathbf{u}. \quad (15)$$

The ideal gas equation of state allows for the calculation of the pressure through the relation,

$$p = (\gamma - 1) \rho e. \quad (16)$$

For mixtures in cells, it is possible to calculate the mixture specific heat ratio γ for two ideal gases using an isobaric closure,

$$\frac{1}{\gamma - 1} = \sum_a \frac{f_a}{\gamma_a - 1}, \quad (17)$$

where γ_a is the specific heat ratio for species a . The equations include a body acceleration vector $\mathbf{g}$, which is further discussed in Section 2.4.

2.2. Numerical methods

Simulations are conducted using FLAMENCO, a multiblock, structured, finite volume solver which uses the Godunov-type method for the evaluation of the inviscid fluxes. The calculation of inviscid flux is performed with the HLLC Riemann solver [17], using the reconstructed values at the cell interfaces from the conserved cell average values. The reconstructed values are obtained using a nominally fifth-order reconstruction [18] which are modified to ensure correct dissipation scaling in the low-Mach number limit [19]. Using a method of lines discretisation,

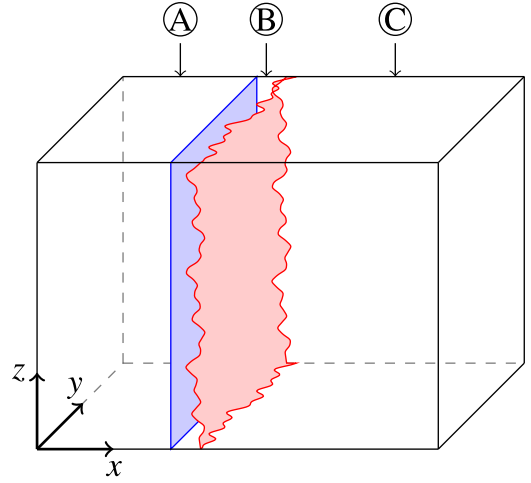


Fig. 2. Initial configuration of the quarter-scale θ -group case. Shockwave is marked in blue, interface is marked in red.

the simulation is integrated forwards in time using the second-order, total-variation diminishing Runge-Kutta scheme [20].

To allow for a moving mesh, the numerical methods are slightly modified. An adjustment to the time integration scheme is needed to follow the geometric conservation law (GCL) [21–23]. The integral form of the GCL is,

$$\frac{\partial}{\partial t} \int_V dV = \int_S \mathbf{W}_S \cdot d\mathbf{S} \quad (18)$$

for a control volume V with boundary S which moves with some velocity $\mathbf{W}_S$. This equation can be derived by simplifying the equations for a stationary, uniform flow. The GCL can be interpreted in two ways. Firstly, the integration scheme used with the moving mesh should conserve the volume of the control volumes. Secondly, that the moving mesh scheme should not induce spurious oscillations for a static flow. For the HLLC solver, the fluxes are modified by accounting for the relative flow velocity, which is the difference between the flow velocity normal to the face and the face velocity from the moving mesh [24].

2.3. Initial conditions

The initial conditions for the simulations are the same as the quarter-scale θ -group case [25]. The simulation domain is $\mathcal{L}_x \times \mathcal{L} \times \mathcal{L} = 2.8\pi \times 2\pi \times 2\pi \text{ m}^3$ and consists of three primary regions as shown in Fig. 2. This central domain is augmented by an extended domain in the x -direction that uses lower-order reconstruction for flux calculations, in order to diffuse any outward travelling fluctuations and minimise the reflections from the supersonic outflow boundary conditions used in the x -direction.

Region A is the shocked heavy fluid and is separated from the unshocked heavy fluid in region B by a shockwave. The shockwave is located at $x = 3 \text{ m}$ and has Mach number of $\text{Ma} = 1.8439$. A perturbed and diffused interface separates regions B and C, where region C contains the unshocked light-fluid. The properties in each region are given by:

$$f_1, \rho, u, p = \begin{cases} 1, 6.57 \text{ kg m}^{-3}, -61.5 \text{ m s}^{-1}, 400 \text{ kPa} & \text{in region A,} \\ 1, 3.00 \text{ kg m}^{-3}, -291.6 \text{ m s}^{-1}, 100 \text{ kPa} & \text{in region B,} \\ 0, 1.00 \text{ kg m}^{-3}, -291.6 \text{ m s}^{-1}, 100 \text{ kPa} & \text{in region C.} \end{cases} \quad (19)$$

These properties give a pre-shock Atwood number of $\mathcal{A}^- = 0.5$, and $\mathcal{A}^+ = 0.487$ for the post-shock configuration. The velocities were designed to ensure that after the shock passage, the interface between the fluids is stationary. Both the light and heavy fluid are ideal gases and

use a specific heat ratio of $\gamma = 5/3$, and molecular weights of 30 g mol^{-1} and 90 g mol^{-1} , respectively. The transition between the two regions is determined by the diffusive width $\delta = L/128$ and the error function profile,

$$f_1(y, z) = \frac{1}{2} \left(1 - \operatorname{erf} \left(\frac{\sqrt{\pi}(x - S(y, z))}{\delta} \right) \right). \quad (20)$$

The interface position $S(y, z)$ is given by

$$S(y, z) = 3.5 + A(y, z) \quad (21)$$

where $A(y, z)$ is the multimode interface perturbation. For the quarter-scale problem, the interface perturbations consist of Fourier modes with wavelengths between $\lambda_{\min} = L/32$ and $\lambda_{\max} = L/16$. A constant power spectrum is used for the perturbation, with full calculations provided in the original paper [25]. For purposes of non-dimensionalisation, the mean wavelength is used for units of length and is given by $\bar{\lambda}_0 = \sqrt{12/7} \lambda_{\min}$. The subscript 0 has been included to denote the initial mean wavelength, as the mean wavelength and domain width change as the system compresses. Velocities are non-dimensionalised by the initial mixing layer growth rate $\dot{W}_0$, where W denotes the integral width,

$$W = \int_0^{L_x} \bar{f}_1(x) \bar{f}_2(x) dx, \quad (22)$$

and $\bar{f}_1(x)$ is the planar average of f_1 over the $y - z$ plane. For the given initial configuration, the initial growth rate is given by $\dot{W}_0 = 12.649 \text{ m s}^{-1}$ [25]. Additionally, the initial cross-sectional area of $A = 4\pi^2 \text{ m}^2$ and the initial average post-shock density $\bar{\rho} = 3.51 \text{ kg m}^{-3}$ are used for non-dimensionalisation. A combination of these variables allow for all other variables to be non-dimensionalised, with the non-dimensionalised time given by

$$\tau = \frac{t \dot{W}_0}{\bar{\lambda}_0}. \quad (23)$$

2.4. Strain application

To evaluate the effect of the compression on the late-time growth, the induced turbulent mixing layer is allowed to evolve until it achieves a state close to self-similarity. At the time of $t = 1.0 \text{ s}$ ($\tau = 49.2$), the velocity field is modified by the inclusion of a linear velocity profile, corresponding to a uniform velocity gradient or strain rate, in the y - and z -directions. To help maintain the uniform strain rate, the boundary conditions in the y - and z -direction are swapped to inviscid moving walls, allowing the ghost cells to maintain the linear velocity profile. The walls, internal mesh points and faces move with the prescribed velocity profile, causing the domain to compress. With strain applied in the transverse (y and z) directions, the transverse expansion factor Λ_T represents the multiplicative growth of the domain,

$$\Lambda_T = \exp \left[\int \bar{S} dt \right], \quad (24)$$

where the mean strain rate $\bar{S}$ under consideration is in the transverse direction, $\bar{S} = \partial \bar{u}_y / \partial y = \partial \bar{u}_z / \partial z$. The strain rate applied is designed to be constant in time and to halve the width of the domain in the transverse directions over a period of application of 0.1 s (4.92τ). The strain rate magnitude is therefore given by,

$$\bar{S} = \frac{\ln(1/2)}{0.1 \text{ s}} \approx -6.93 \text{ s}^{-1}. \quad (25)$$

As strain is only applied in the y - and z -directions, the fluid experiences net compression. The compression increases the final density and pressure of the fluids by a factor of 4 and 4^{γ} , respectively. By applying the strain from $t = 1 \text{ s}$ to $t = 1.1 \text{ s}$, a large period of time before and after allows any transient behaviours from acoustic waves to disappear, allowing the mixing layer to tend towards, or achieve, a self-similar state. In practical applications, the ratio of the strain timescale to the turbulent timescale will vary depending upon the initial conditions and

Table 1

Computational meshes used for the simulations, listed in order of increasing resolution. Cell sizes listed are for the initial mesh, before the compression phase.

Mesh	N_x	N_y	N_z	Δx_0 (m)	$\Delta y_0 = \Delta z_0$ (m)
A	180	128	128	0.0489	0.0491
B	360	256	256	0.0244	0.0245
C	720	512	512	0.0122	0.0123

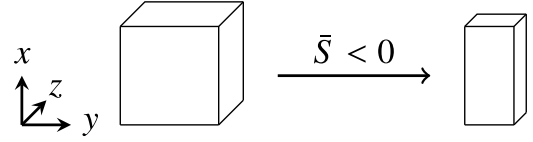


Fig. 3. Deformation of the computational cells during the compression phase of the simulations.

throughout the simulation [8]. For the conditions provided, the system is expected to be strain dominated during the implosion process.

Using a constant strain rate profile requires a potential forcing to be used to accelerate the flow without a pressure differential. The body force for the given problem with strain in the y - and z -directions is not uniform in space, given by

$$\mathbf{g} = [0, \bar{S}^2 y, \bar{S}^2 z]^T. \quad (26)$$

At $t = 1.1 \text{ s}$ ($\tau = 54.2$), the uniform strain rate and potential forcing is removed, and the boundary conditions are reverted to periodic. The simulation is conducted until $t = 2.5 \text{ s}$ ($\tau = 123$), allowing the compressed turbulent mixing layer to evolve in the absence of the strain.

Simulations are conducted using three different meshes with specifications provided in Table 1. Simulations are conducted on each mesh for the base case (no strain applied) and for the strained case (strain application for $\tau \in [1, 1.1] \text{ s}$).

The meshes are designed to begin with initially isotropic cell sizes and follow the typical refinement process used for mesh convergence, with mesh C being equivalent to the standard mesh size used for the original investigation of the quarter-scale θ -group case [25]. With the application of the strain rates and moving mesh, the computational cells are deformed during the simulation. Due to the applied compression, the initially cubic cells become anisotropic and columnar, as demonstrated in Fig. 3.

For ILES, the simulations rely on numerical dissipation to remove energy from the flow, where the dissipation rate is determined by the energy transfer rate from the large scales. The scales at which numerical dissipation occurs and the resulting turbulent kinetic energy spectrum for the flow is dependent upon the numerical scheme and the grid resolution used. The usage of anisotropic cell sizes can affect the directional dissipation rates as well as the types of vortical structures that can be resolved. As the initial mesh spacing is refined, these impacts move to smaller scales with diminishing influence, and still allow the large scale features to be resolved, and this is the approach undertaken here.

3. Results

3.1. Qualitative comparison

The flow field prior to the strain application at $\tau = 49.2$ is shown in Fig. 4, contoured by the volume fraction of the mixing layer. The mixing layer is observed to have turbulent features, with a range of scales present, and there are some visible structures penetrating into the pure fluid regions above and below the mixing layer. In Fig. 5 the flow field is visualised at $\tau = 54.1$ which corresponds to the time at which the strained cases have reached the final compression and the strain rate is removed. Both the base and the strained flow fields are presented, with negligible differences in the base case as compared to Fig. 4. The

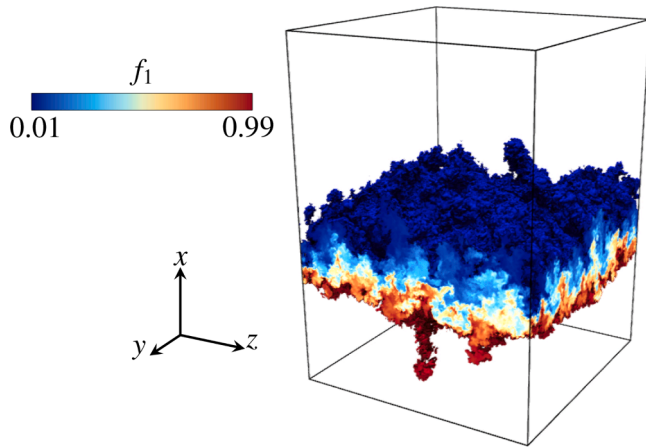


Fig. 4. Volume fraction contour of the mixing layer for mesh C at $\tau = 49.2$, bound by the $f_1 = 0.01$ and 0.99 isosurfaces.

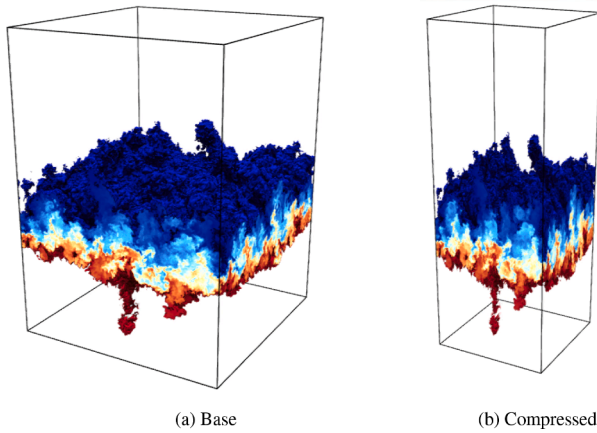


Fig. 5. Volume fraction contour of the mixing layer for mesh C at $\tau = 54.1$, bound by the $f_1 = 0.01$ and 0.99 isosurfaces.

strained domain in contrast appears visibly compressed in the y - and z -direction. This is observed both by the reduction in size of the bounding box which shows the domain, but also in the structures within the mixing layer. The structures have not had time to adapt due to the rapid compression, causing the structures to appear thinner relative to their penetration through the mixing layer. The mixing layer continues to evolve for both cases without any applied strain, and by the final time of $\tau = 123$ the internal features of the mixing layer for the strained case no longer appear as anisotropic, as shown in Fig. 6. For both cases, the visualisations no longer show some of the penetrating structures outside of the core mixing layer due to species diffusion, meaning the structures are no longer captured within the volume fraction limits used for the visualisation. Between the strained and unstrained mixing layer, the mixing layers do not display any immediate visual differences regarding the internal structures and turbulence.

3.2. Mixing layer growth

Quantifying the penetration of fluids and the growth of the turbulent mixing layer is important for design and/or modelling purposes for ICF and supernova. The key metric used for the growth of the mixing layer is the integral width, as defined in Eq. (22). The integral width is calculated with the mean volume fraction profile, causing the measurement to be less sensitive to statistical fluctuations in the development of the mixing layer as compared to measurements which use cut-off limits. Without any explicit dependence on the instantaneous/fluctuation

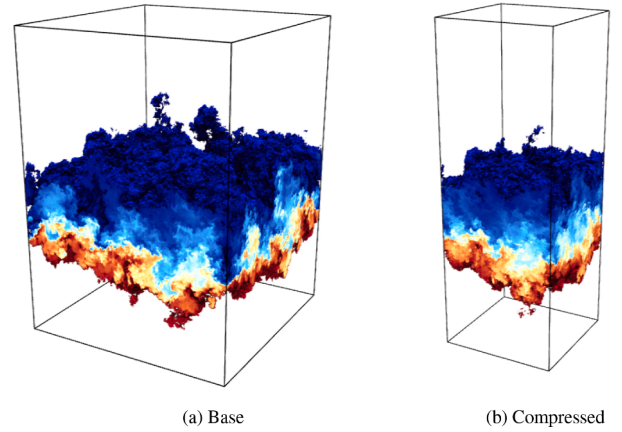


Fig. 6. Volume fraction contour of the mixing layer for mesh C at $\tau = 123$, bound by the $f_1 = 0.01$ and 0.99 isosurfaces.

values, it does not account for the mixedness within the mixing layer. The integral width for the cases are presented in Fig. 7, showing to be converged for mesh C, with minimal variation from mesh B. During the period of strain application, as marked by the grey region of the plot, there is no visible difference between the unstrained and strained cases. The effect of the bulk compression manifests over a longer time period than the compression duration. The compressed cases grow at a reduced rate compared to the base cases.

To quantify the change in growth rate and the effect of the compression, it is possible to evaluate the corresponding value of θ , assuming the mixing layer growth is self-similar. A time-series for the value of θ can be calculated by using the drag equation for RMI in the self-similar regime,

$$\dot{V} = -C_D \frac{V^2}{W}, \quad (27)$$

where the overdot denotes the time derivative, and V is the growth rate of the mixing layer, as defined by the relation $V = \dot{W}$. For the solution of the form $W \propto t^\theta$, the value of the θ can be obtained by using the relation $\theta^{-1} = 1 + C_D$. With the measurement of the integral width, the derivatives can be numerically calculated and used to find θ . These values are presented in Fig. 8 for both cases. Due to the noise during the strain application, the results for the compressed cases are only presented for after the period of compression. During the period of compression, the base cases for meshes B and C have a value of $\theta = 0.3$, however this value slightly decreases as the simulation progresses. The results in the θ -group collaboration [25] show that mesh C continues to decrease towards a value of 0.25 by the time of $\tau = 250$, which is beyond the range of the simulations conducted for the current investigation. In contrast, the compressed cases show a smaller value of the θ after the period of compression, with values just below $\theta = 0.1$ visible. The value of the θ slightly increases with time, reaching $\theta = 0.14$, which is still around half the value of the base cases. The increasing trend for θ implies the compression effects are only transient, and the growth rate could eventually recover to the original value, albeit on a significantly longer time scale than simulated here. The eddy turnover period of the large scales, calculated by $\tau_{\text{eddy}} = W/V$, increases linearly in time for unstrained RMI, making it difficult to achieve a large number of eddy turnovers in a simulation. For mesh C, the simulation time after the strain application amounts to 0.24 eddy turnovers for the base case and 0.16 turnovers for the compression case. It is reasonable to expect that by one full turnover the compressed flow may reattain the characteristics of the base case. Modelling of the growth rate should therefore account for not only the compression which affects the mixing layer growth, but also the slow decay of the compression effect.

For the conditions under consideration, taking $\theta = 0.3$ for the base case and $\Lambda_T = 0.5$, Eq. (8) predicts $\hat{\theta} \approx 0.176$ for the modified growth

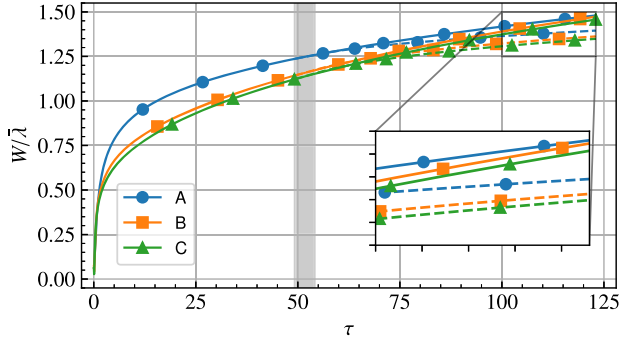


Fig. 7. Temporal evolution of the integral width of the mixing layer. Solid lines represent the base cases, dashed lines represent the compressed cases. The grey region indicates the period of applied strain.

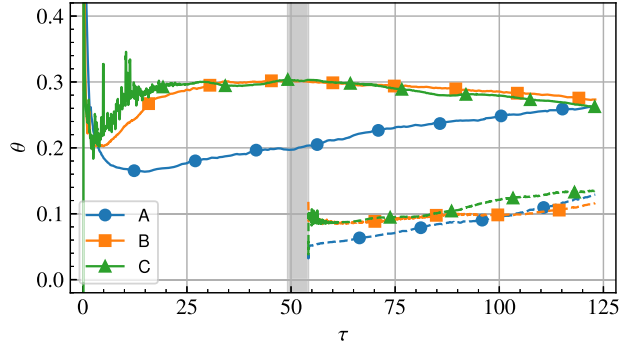


Fig. 8. Temporal evolution of the evaluated growth rate parameter θ from the integral width data. Solid lines represent the base cases, dashed lines represent the compressed cases. The grey region indicates the period of applied strain.

rate. This value is larger than the values observed in Fig. 7 for the compressed cases but as noted the post-strain growth rate is increasing. The growth rate may not exactly match immediately post-strain, however it is an improvement over the models which do not account for a change in θ . Given the increasing eddy turnover times for RMI, should the growth rate continue to rise it may not return to the original θ in physical conditions, such that the improved model would perform better than the original.

Calculating the power-law growth rate using the derivatives of the width measurements is not always possible. Noise from experimental measurements will be compounded in the derivatives, potentially resulting in incoherent results for θ . An alternative method is to curve fit the data, which is typically done with a non-linear least squares regression. This process was performed using the *SciPy* package and simulation data in the domain of $\tau \in [73, 123]$, where the time after the compression period has been excluded to avoid any transient behaviours and instead capture the steady state characteristics. The results are listed in Table 2, alongside the theoretical values of $\hat{\theta}$ that are predicted by Eq. (8) for the compression cases. With this methodology, the values of θ are smaller than the base cases, in-line with the results in Fig. 8. The values are still lower than the theory, however this method does not give any indication for how θ changes in time.

A second measure used to evaluate the mixing layer growth is the mixed mass. The mixed mass $\mathcal{M}$ is calculated by integrating over the flow domain,

$$\mathcal{M} = \int \rho Y_1 Y_2 dV. \quad (28)$$

It is possible to derive the transport equation for the mixed mass, showing that it increases due to diffusion, which can be either physical or

Table 2

Values of θ from curve-fitting the integral width data. Compressed theory values are calculated from Eq. (8) and the base value.

Mesh	Base	Compressed	Compressed Theory
A	0.245	0.091	0.157
B	0.286	0.098	0.193
C	0.279	0.112	0.186

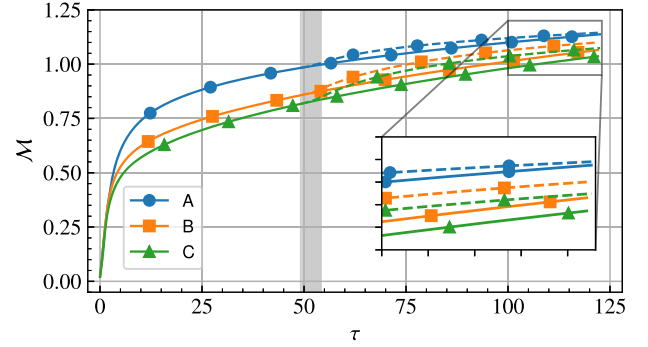


Fig. 9. Temporal evolution of the mixed mass of the mixing layer. Solid lines represent the base cases, dashed lines represent the compressed cases. The grey region indicates the period of applied strain.

numerical [26,27]. As the mixed mass does not depend upon the mean species profile, but instead the instantaneous values, it is affected by the mixedness of the mixing layer. The effect of this is observed in the Fig. 9 for the compression cases. The base cases maintain a steady power-law growth as observed for the integral width, however the compressed cases behave differently. After the strain application, the mixed mass for the compression cases increase relative to the base cases. As the simulations progress, this difference diminishes. These results are the combination of the mixing layer becoming more mixed after compression, but growing at a slower rate.

3.3. Mixedness

The mixedness of the mixing layer is a useful metric for measuring convergence and demonstrating self-similarity of the mixing layer. It is also important to quantify the mixedness to account for the difference in the behaviour between the integral width and mixed mass that is observed for the compressed cases. The first quantity used to measure the mixedness is the molecular mixing fraction,

$$\Theta = \frac{\int \bar{f}_1 \bar{f}_2 dV}{\int \bar{f}_1 \bar{f}_2 dV} = 1 + \frac{\int \bar{f}_1' \bar{f}_2' dV}{\int \bar{f}_1 \bar{f}_2 dV}. \quad (29)$$

The integral of the fluctuation product is negative for binary mixtures, as the increase of the volume fraction for one species will be correlated with a decrease in the other. The molecular mixing fraction is constrained to the range of $\Theta \in [0, 1]$, where $\Theta = 0$ represents a mixing layer with the fluids segregated in each $y - z$ plane, whilst for $\Theta = 1$ the fluid mixture throughout the plane is homogenous. The simulation values in Fig. 10 show the base cases converge towards a steady value around $\Theta = 0.8$ for mesh B and C. The compression cases begin to deviate during the strain application period and continue to increase after the strain is removed. The behaviour of Θ for the compressed cases appear steady, reaching a value around $\Theta = 0.9$ at the final simulation time for mesh B and C. This increase in mixing following the compression is consistent with the increase observed for the mixed mass. Further, it is expected that a slower growing mixing layer attains a greater level of mixing. If the value of θ continues to increase for the mixing layer post-compression, the corresponding steady state value of Θ should decrease.

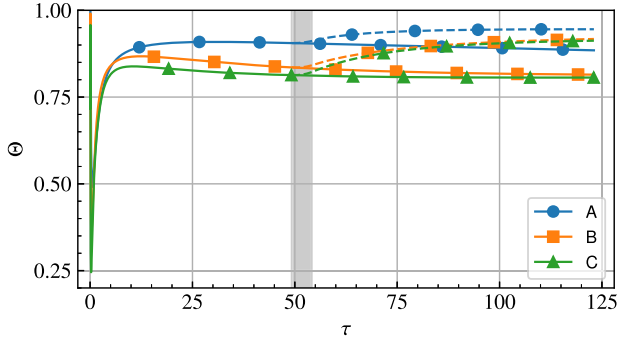


Fig. 10. Temporal evolution of the molecular mixing fraction for the mixing layer. Solid lines indicate the base cases, dashed lines indicate the compressed cases. The grey region indicates the period of applied strain.

3.4. Turbulent kinetic energy

Once a turbulent kinetic energy cascade is established, it is necessary to resolve a sufficiently wide bandwidth of length scales and velocities of the flow necessary to appropriately separate the large and dissipative scales. Using ILES without any physical viscosity, thereby simulating the high-Reynolds number limit, the turbulent kinetic energy spectrum will not reach the physical dissipation scales. Instead, the algorithm aims to maximise the bandwidth between the large scale and scales where numerical dissipation is active. With the contribution to the total turbulent kinetic energy decreasing with an increase in wavenumber, there should be convergence in the total amount of turbulent kinetic energy despite the growing inertial range. To evaluate the turbulent kinetic energy, the fluctuating components are calculated by removing the mean flow. This is done by removing the velocity profile which corresponds to the applied strain rate, and removing the Favre-averaged component. The Favre-average applies a density weighting to the averaging process done over the $y - z$ planes which are expected to be statistically homogenous. The resulting velocity fluctuation will have a density weighted average of zero. With the strain rates removed from u_i , the fluctuating velocity components are

$$u_i'' = u_i - \frac{\rho u_i}{\bar{\rho}} \quad (30)$$

For the evaluation of the turbulent kinetic energy in the domain, the components of the energy can be separated by direction, with TKX representing the x -direction. With the statistical homogeneity in the $y - z$ plane, the components are averaged to give $TKYZ$. The two components used are:

$$TKX = \int \frac{1}{2} \rho u_1'' u_1'' dV, \quad (31)$$

$$TKYZ = \int \rho \frac{u_2'' u_2'' + u_3'' u_3''}{4} dV. \quad (32)$$

To calculate the total turbulent kinetic energy, the terms are summed,

$$TKE = TKX + 2 \times TKYZ. \quad (33)$$

The presented value for TKE in Fig. 11 show reasonable convergence between mesh B and C for the simulations. The base cases are linear in the log-log plot, as expected for a power law decay rate of the turbulent kinetic energy. The compressed cases deviate from the base cases for two reasons. Firstly, during the strain application process there is a production of turbulent kinetic energy from the velocity gradients. The production of energy is much larger than the dissipation rate, causing the TKE to increase. This is expected in the rapid-distortion limit, which occurs if the strain timescale is much smaller than the turbulence timescale [28,29]. Swapping to the notation used for rapid-distortion theory, $k = TKX$, and estimating the dissipation rate as $\epsilon = -dk/dt$, it

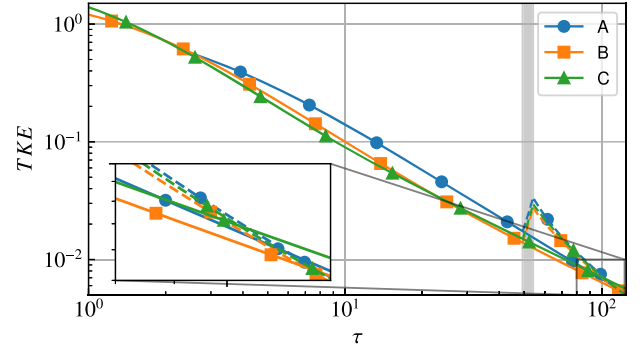


Fig. 11. Temporal evolution of the total fluctuating kinetic energy. Solid lines indicate the base cases, dashed lines indicate the compressed cases. The grey region indicates the period of applied strain.

is possible to calculate the non-dimensional strain rate:

$$\frac{\bar{S}k}{\epsilon} \simeq 6.5. \quad (34)$$

Whilst this value is not ‘much greater’ than one and the system may not qualify for rapid-distortion, it does show that strain effects are dominant during the compression phase. If the non-dimensional strain is instead quantified using the large scale parameters from the integral width, then

$$\frac{\bar{S}W}{V} \simeq 24.7, \quad (35)$$

which is almost four times larger. An increase is reasonable, as it is expected that the eddy turnover time decreases as the wavenumber increases [30], and so the turbulent kinetic energy timescale should be shorter than the timescale of the mixing layer/large scales.

After the strain is removed, the compressed cases have larger TKE but this is counteracted by an increase in the dissipation rate. An increase in the dissipation rate can be obtained by the compression of the turbulent length scale, thereby increasing turbulent dissipation [9]. Further, the decay rate of the turbulent kinetic energy for RMI can be well modelled according to $t^{3\theta-2}$ [31]. With the smaller values of θ for the compressed cases, the decay rate is expected to become more negative, as observed in Fig. 11. By the end of simulation time, the compressed cases’ values of TKE have dropped below the values for the base cases. Even though the TKE values are initially higher, with much larger dissipation rates the eddy turnover time k/ϵ has decreased for the compressed cases. Following the strain application period, the compressed cases have completed almost 1.8 eddy turnovers as compared to 0.88 for the base cases over the same time. This is in contrast to the large scales, for which the base cases achieved more eddy turnovers than the compressed cases.

The shear production from the compressive strain rates primarily adds energy in the transverse direction, with some of the energy being redirected into the x -direction. The ratio of the directional energy components can be calculated with

$$TKR = \frac{TKX}{TKYZ}. \quad (36)$$

In Fig. 12, the base cases tend towards an asymptotic non-unity value, representing persistent anisotropy. The compression cases show a strong decrease in TKR during the compression phase due to shear production for $TKYZ$. The post-strain results for the compressed cases are not well converged, and increase in the final value for TKR . Mesh C approaches a value of isotropy, but this is likely a coincidence. It is expected that once the shear production is halted, there will continue to be some re-distribution of the turbulent kinetic energy between the components. As the TKR is not properly converged, it is inconclusive as to what value the ratio is tending towards at late time. It is evident however, that the flow is not immediately rebounding to the base asymptotic value, and

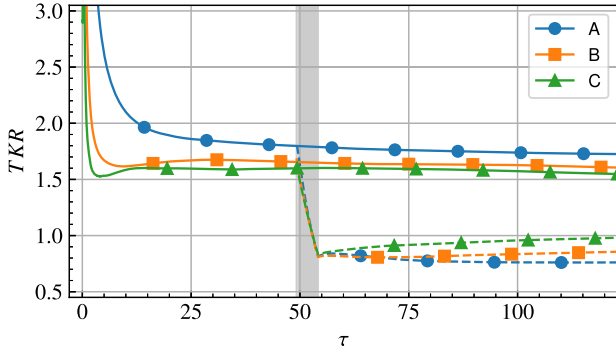


Fig. 12. Temporal evolution of the fluctuating kinetic energy anisotropy. Solid lines represent the base cases, dashed lines represent the compressed cases. The grey region indicates the period of applied strain.

the new asymptotic value has shifted in the direction of the added energy. This is similar to re-shocked RMI, which initially deposits energy in the x -direction, is observed to achieve a mixing layer with a larger anisotropy ratio [32]. Likewise, direct numerical simulations of shock-turbulence interaction of isotropic turbulence have shown the flow field to not return to isotropy, with any return-to-isotropy timescale much larger than the turbulent decay timescale [33].

3.4.1. Spectra

The turbulent kinetic energy spectrum in the inertial range can vary from the trademark $-5/3$ slope predicted by Kolmogorov [34]. For RMI, it is common for the spectrum to follow a $-3/2$ slope [31,35]. To analyse the distribution of the turbulent kinetic energy, the power spectra of the velocity components is calculated. The two-dimensional power spectra are calculated for a series of planes. The two-dimensional spectra are converted to radial one-dimensional power spectra for a single wavenumber,

$$\kappa = \sqrt{\kappa_y^2 + \kappa_z^2}. \quad (37)$$

To minimise noise in the signal, the 1D radial power spectra for $y-z$ planes with at least 90 % of the maximum planar turbulent kinetic energy are averaged. The out-of-plane and in-plane components are represented by $E_x(\kappa)$ and $E_{yz}(\kappa)$, respectively. Due to the variable density, the power spectrum includes a density weighting,

$$\psi_x = \sqrt{\rho} u_1 \quad (38a)$$

$$E_x(\kappa) = \hat{\psi}_x \cdot \hat{\psi}_x^\dagger \quad (38b)$$

$$\psi_{yz} = \sqrt{\rho} [u_2, u_3]^T \quad (38c)$$

$$E_{yz}(\kappa) = \frac{1}{2} \hat{\psi}_{yz} \cdot \hat{\psi}_{yz}^\dagger, \quad (38d)$$

where $\hat{\psi}$ is the Fourier transform of ψ , and $\hat{\psi}^\dagger$ is the complex conjugate. The radial power spectra are presented in Figs. 13 and 14 for the flow at $\tau = 123$. The results for mesh A have been removed to improve the visibility of the higher quality simulations. The wavenumber on the x -axis of the plots has not been non-dimensionalised, allowing it to represent the physical wavenumber. For the base cases with a domain width of 2π , the wavenumber increment is $\Delta\kappa = 1$. The compressed cases have half the domain width, which allows it to resolve twice the maximum wavenumber, but only the even modes ($\Delta\kappa = 2$). It can be observed in both plots that the compressed cases extend to twice the wavenumber of the base case for the given resolution.

Included in the figures is the expected power law scaling of the inertial range, with $\kappa^{-3/2}$ scaling. From $\kappa = 10$ onwards, the simulation cases show the presence of an inertial range with a reasonable alignment with the specified scaling. For the base cases, the inertial range is converged and is terminated by an up-tick in the energy for the highest resolved wavenumbers, known as an energy pile-up.

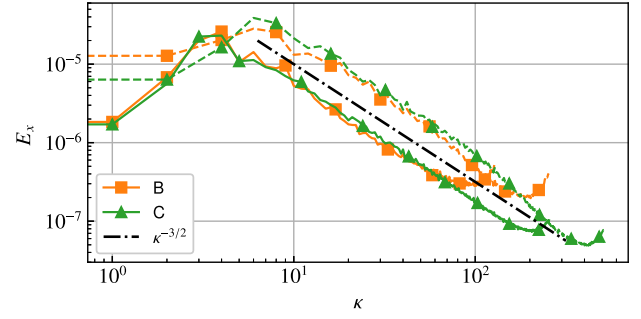


Fig. 13. One-dimensional radial power spectra for the variable density kinetic energy in the x -direction at the centre-plane at the final time of $\tau = 123$. Solid lines represent the base cases, dashed lines represent the compressed cases.

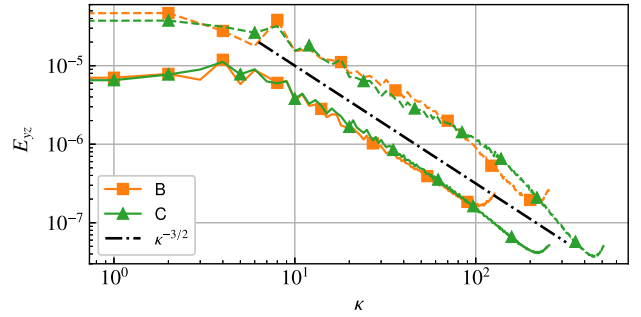


Fig. 14. One-dimensional radial power spectra for the variable density kinetic energy in the y - and z -direction at the centre-plane at the final time of $\tau = 123$. Solid lines represent the base cases, dashed lines represent the compressed cases.

The compressed cases are not quite as well converged as the base cases, with a larger separation in the energy at the large scales for E_x . Rather than showing a single consistent trend for the majority of length scales, the compressed cases transition to a steeper decrease in energy at the higher wavenumbers before the energy pile-up at the highest wavenumber. The transition point, where the spectrum swaps to the steeper decrease, moves to a higher wavenumber as the resolution improves. This spectral behaviour can be attributed to the anisotropic mesh that is the result of the compression phase, as shown in Fig. 3. Taking the transition points for E_{yz} as $\kappa = 80$ for mesh B, and $\kappa = 150$ for mesh C corresponds to a length scale that is between six and seven cells across in the transverse direction, and half that in the x -direction. The resolution of any vortices at such small lengths is difficult, and many LES codes which model RMI [25] or RTI [36] display a stronger drop off in resolved energy at the high wavenumbers.

To compare how the velocity field has been affected by the compression process, the velocity field in the transverse direction can be decomposed into the dilatational and solenoidal components of the flow,

$$\psi_{yz} = \nabla\phi + \nabla \times \xi, \quad (39)$$

for some scalar field ϕ and vector field ξ . Using the Fourier transformed velocity field, the turbulent kinetic energy in each of the decomposed fields can be calculated according to,

$$\widehat{\nabla\phi} = \frac{\kappa \cdot \hat{\psi}_{yz}}{|\kappa|^2} \kappa \quad (40a)$$

$$E_{yz}^{(s)} = \widehat{\nabla\phi} \cdot \widehat{\nabla\phi}^\dagger \quad (40b)$$

$$\widehat{\nabla \times \xi} = \hat{\psi}_{yz} - \widehat{\nabla\phi} \quad (40c)$$

$$E_{yz}^{(d)} = \widehat{\nabla \times \xi} \cdot \widehat{\nabla \times \xi}^\dagger \quad (40d)$$

For direct numerical simulations of variable density, it is also possible to separate out the velocity field component that corresponds to

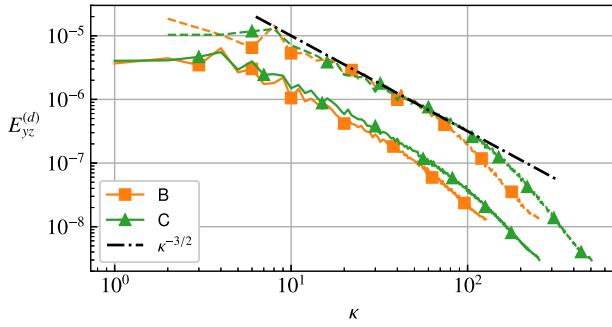


Fig. 15. One-dimensional radial power spectra for the dilatational variable density kinetic energy in the y - and z -direction at the centre-plane at the final time of $\tau = 123$. Solid lines represent the base cases, dashed lines represent the compressed cases.

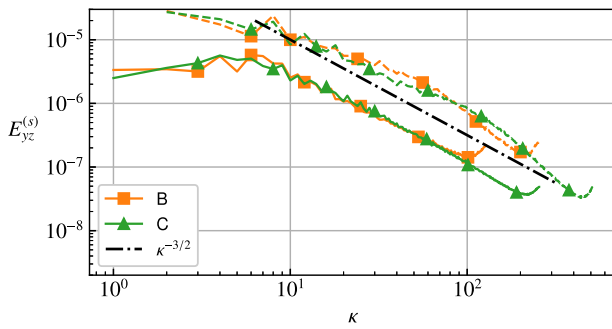


Fig. 16. One-dimensional radial power spectra for the solenoidal variable density kinetic energy in the y - and z -direction at the centre-plane at the final time of $\tau = 123$. Solid lines represent the base cases, dashed lines represent the compressed cases.

the diffusive mixing of the fluids [37]. This is not possible for ILES, without quantifying the numerical diffusion. The energy for the dilatational and solenoidal field is presented in Figs. 15 and 16 as a function of the radial wavenumber for the final simulation time. The black dot-dash line included in the figures is in the same location as Fig. 14 and still represents the $-5/3$ slope. The dilatational components and the solenoidal component of the base simulations are aligned with the $-5/3$ slope, while the compressed solenoidal profile is less converged. The compression simulation results show larger energy in both of the components of the flow, and the dilatational energy is lower than the solenoidal energy for both cases. This is expected as the turbulent Mach number and compressibility effects being small for RMI at late-time, but still has contributions to the dilatational energy due to the density differences in the flow [38]. The energy pile-up observed for the total spectrum at the highest wavenumbers is only visible for the solenoidal energy, while the dilatational field instead shows a steeper decrease in energy.

4. Conclusion

We have investigated the late-time evolution of a Richtmyer-Meshkov turbulent mixing layer subjected to a period of bulk compressive strain, using high-resolution implicit LES. By comparing against an unstrained baseline, we found that compression has a pronounced suppressing effect on the mixing layer's growth rate. Quantitatively, the instantaneous growth exponent θ reduces by more than 50 % following the compression event - from an initial $\theta \approx 0.25$ -0.30 in the self-similar unstrained flow down to $\theta \approx 0.10$ -0.14 in the compressed cases. The effect of the imposed strain is not just during the application of the strain it-

self but is also sustained over a long period following its removal: by the end of our simulations the compressed mixing layer had not re-attained the original growth rate, indicating that the effects persist over many eddy-turnover times.

Accompanying this reduced growth, the mixing characteristics of the layer were notably altered by compression. In all cases, the mixed mass M increased more rapidly when the layer was compressed, immediately overshooting the mixed mass of the unstrained case once the strain was removed. This indicates that the compression caused additional interpenetration of the fluids (enhancing small-scale mixing) even as the overall growth in the mixing width was curtailed. Consistent with this, the molecular mixing fraction Θ rose to higher values under compression: the compressed layers reached $\Theta \approx 0.9$ by late time, compared to about 0.8 in the unstrained case, indicating that the compressed mixing layer became more homogeneously mixed (closer to a uniform mixture) than its unstrained counterpart.

The turbulent kinetic energy (TKE) in the flow was also strongly impacted by the compression. During the brief application of strain, there was a surge in TKE in the compressed cases, reflecting rapid work done on the turbulence by the imposed velocity gradients. This is analogous to the classical rapid distortion effect: the compression added energy to the fluctuating motions faster than it could be dissipated, leading to a transient increase in total TKE. However, this injected turbulent energy was quickly dissipated after the strain ended, such that by the end of the simulation the compressed case had lower overall TKE than the base case. This is because the compression reduces the turbulent length scales (effectively shifting energy to smaller eddies), which in turn raises the dissipation rate and which is also consistent with a smaller exponent θ . We also observed a redistribution of TKE among components: the shear production associated with compression fed energy primarily into fluctuations in the transverse directions, temporarily driving the turbulence toward isotropy. After the strain was removed, some re-equilibration occurred, but the compressed layer maintained a different anisotropy ratio than the baseline case.

In summary, imposing a brief transverse compression on a turbulent mixing layer has the net effect of creating a thinner, more thoroughly mixed, and less energetic layer in the long term. The growth rate reduction and enhanced mixedness observed in our simulations are in line with theoretical predictions that account for geometric strain. In particular, the $\hat{\theta}$ model introduced in the introduction successfully captures the direction and general magnitude of the growth-rate modification due to compression. The measured post-compression exponent in our simulations ($\hat{\theta} \sim 0.1$ -0.15) is slightly lower than the simple model estimate ($\hat{\theta} \sim 0.15$ -0.19 for a 50 % domain compression), but the model correctly predicts a significant drop in θ and thus offers a useful first-order correction for compression effects. This agreement lends support to incorporating strain-based adjustments in models of RMI-induced turbulence [7]. Such improvements are important for extending turbulence modelling of RMI to the convergent geometries encountered in imploding ICF capsules and astrophysical flows. Future studies will build on this framework to examine longer-term recovery after compression, as well as investigate the interplay of simultaneous axial and transverse strains.

CRedit authorship contribution statement

Bradley Pascoe: Writing – review & editing, Writing – original draft, Software, Resources, Methodology, Formal analysis, Conceptualization; **Michael Groom:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization; **Ben Thornber:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This research was supported by the Australian Government's National Collaborative Research Infrastructure Strategy (NCRIS), with access to computational resources provided by the Setonix supercomputer through the National Computational Merit Allocation Scheme. The authors thank EPSRC for the computational time made available on the UK supercomputing facility ARCHER/ARCHER2 via the UK Turbulence Consortium (Grant No. EP/R029326/1).

References

- [1] Y. Zhou, Rayleigh–Taylor and Richtmyer–Meshkov instability induced flow, turbulence, and mixing. I, *Phys. Rep.* 720–722 (2017) 1–136.
- [2] Y. Zhou, Rayleigh–Taylor and Richtmyer–Meshkov instability induced flow, turbulence, and mixing. II, *Phys. Rep.* 723–725 (2017) 1–160.
- [3] Y. Zhou, R.J.R. Williams, P. Ramaprabhu, M. Groom, B. Thornber, A. Hillier, W. Mostert, B. Rollin, S. Balachandrar, P.D. Powell, A. Mahalov, N. Attal, Rayleigh–Taylor and Richtmyer–Meshkov instabilities: a journey through scales, *Phys. D* 423 (2021) 132838.
- [4] Y. Zhou, *Hydrodynamic Instabilities and Turbulence: Rayleigh–Taylor, Richtmyer–Meshkov, and Kelvin–Helmholtz Mixing*, Cambridge University Press, 2024.
- [5] Y. Zhou, J.D. Sadler, O.A. Hurricane, Instabilities and mixing in inertial confinement fusion, *Annu. Rev. Fluid Mech.* 57 (Volume 57, 2025) (2025) 197–225.
- [6] M. El Rafei, B. Thornber, Turbulence statistics and transport in compressible mixing driven by spherical implosions with narrowband and broadband initial perturbations, *Phys. Rev. Fluids* 9 (3) (2024) 034501.
- [7] B. Pascoe, M. Groom, B. Thornber, Turbulence modeling of mixing layers under anisotropic strain, *Phys. Rev. Fluids* 10 (2025) 064609. <https://doi.org/10.1103/PhysRevFluids.10.064609>
- [8] B. Pascoe, M. Groom, D.L. Youngs, B. Thornber, Impact of axial strain on linear, transitional and self-similar turbulent mixing layers, *J. Fluid Mech.* 999 (2024) A5.
- [9] B. Pascoe, M. Groom, D.L. Youngs, B. Thornber, Impact of transverse strain on linear, transitional and self-similar turbulent mixing layers, (2025). <https://arxiv.org/abs/2502.13444>
- [10] D.L. Youngs, B. Thornber, Buoyancy–drag modelling of bubble and spike distances for single-shock Richtmyer–Meshkov mixing, *Phys. D* 410 (2020) 132517.
- [11] A.R. Miles, The blast-wave-driven instability as a vehicle for understanding supernova explosion structure, *Astrophys. J.* 696 (2009) 498–514.
- [12] M. El Rafei, B. Thornber, Numerical study and Buoyancy–drag modeling of bubble and spike distances in three-dimensional spherical implosions, *Phys. Fluids* 32 (12) (2020) 124107.
- [13] J. Lindl, O. Landen, J. Edwards, E. Moses, NIC Team, Review of the national ignition campaign 2009–2012, *Phys. Plasmas* 21 (2) (2014) 020501.
- [14] R.E. Olson, R.J. Leeper, J.L. Kline, A.B. Zylstra, S.A. Yi, J. Biener, T. Braun, B.J. Kozioziemski, J.D. Sater, P.A. Bradley, R.R. Peterson, B.M. Haines, L. Yin, L.F. Berzak Hopkins, N.B. Meezan, C. Walters, M.M. Biener, C. Kong, J.W. Crippen, G.A. Kyrila, R.C. Shah, H.W. Herrmann, D.C. Wilson, A.V. Hamza, A. Nikroo, S.H. Batha, First liquid layer inertial confinement fusion implosions at the national ignition facility, *Phys. Rev. Lett.* 117 (24) (2016) 245001.
- [15] M. Brouillette, R. Bonazza, Experiments on the Richtmyer–Meshkov instability: wall effects and wave phenomena, *Phys. Fluids* 11 (5) (1999) 1127–1142. https://pubs.aip.org/aip/pof/article-pdf/11/5/1127/19301513/1127_1_online.pdf. <https://doi.org/10.1063/1.869983>
- [16] G. Allaire, S. Clerc, S. Kokh, A five-equation model for the simulation of interfaces between compressible fluids, *J. Comput. Phys.* 181 (2) (2002) 577–616.
- [17] E.F. Toro, M. Spruce, W. Speares, Restoration of the contact surface in the HLL–Riemann solver, *Shock Waves* 4 (1) (1994) 25–34.
- [18] K.H. Kim, C. Kim, Accurate, efficient and monotonic numerical methods for multi-dimensional compressible flows: Part II: multi-dimensional limiting process, *J. Comput. Phys.* 208 (2) (2005) 570–615.
- [19] B. Thornber, A. Mosedale, D. Drikakis, D. Youngs, R.J.R. Williams, An improved reconstruction method for compressible flows with low mach number features, *J. Comput. Phys.* 227 (10) (2008) 4873–4894.
- [20] R.J. Spiteri, S.J. Ruuth, A new class of optimal high-order strong-stability-preserving time discretization methods, *SIAM J. Numer. Anal.* 40 (2) (2002) 469–491.
- [21] D.J. Mavriplis, Z. Yang, Construction of the discrete geometric conservation law for high-order time-accurate simulations on dynamic meshes, *J. Comput. Phys.* 213 (2) (2006) 557–573.
- [22] J.Y. Trépanier, M. Reggio, H. Zhang, R. Camarero, A finite-volume method for the Euler equations on arbitrary Lagrangian–Eulerian grids, *Comput. Fluids* 20 (4) (1991) 399–409.
- [23] C. Farhat, P. Geuzaine, C. Grandmont, The discrete geometric conservation law and the nonlinear stability of ALE schemes for the solution of flow problems on moving grids, *J. Comput. Phys.* 174 (2) (2001) 669–694.
- [24] H. Luo, J.D. Baum, R. Löhner, On the computation of multi-material flows using ALE formulation, *J. Comput. Phys.* 194 (1) (2004) 304–328.
- [25] B. Thornber, J. Griffond, O. Poujade, N. Attal, H. Varshochi, P. Bigdelou, P. Ramaprabhu, B. Olson, J. Greenough, Y. Zhou, O. Schilling, K.A. Garside, R.J.R. Williams, C.A. Batha, P.A. Kuchugov, M.E. Ladonkina, V.F. Tishkin, N.V. Zmitrenko, V.B. Rozanov, D.L. Youngs, Late-time growth rate, mixing, and anisotropy in the multimode narrowband Richtmyer–Meshkov instability: the θ -group collaboration, *Phys. Fluids* 29 (10) (2017) 105107.
- [26] Y. Zhou, W.H. Cabot, B. Thornber, Asymptotic behavior of the mixed mass in Rayleigh–Taylor and Richtmyer–Meshkov instability induced flows, *Phys. Plasmas* 23 (5) (2016) 052712.
- [27] Y. Zhou, M. Groom, B. Thornber, Dependence of enstrophy transport and mixed mass on dimensionality and initial conditions in the Richtmyer–Meshkov instability induced flows, *J. Fluids Eng.* 142 (12) (2020).
- [28] J.C.R. Hunt, D.J. Carruthers, Rapid distortion theory and the ‘Problems’ of turbulence, *J. Fluid Mech.* 212 (1990) 497–532.
- [29] S.B. Pope, *Turbulent Flows*, Cambridge University Press, 2000.
- [30] B. Thornber, Y. Zhou, Energy transfer in the Richtmyer–Meshkov instability, *Phys. Rev. E* 86 (5) (2012) 056302.
- [31] B. Thornber, D. Drikakis, D.L. Youngs, R.J.R. Williams, The influence of initial conditions on turbulent mixing due to Richtmyer–Meshkov instability, *J. Fluid Mech.* 654 (2010) 99–139.
- [32] B. Thornber, D. Drikakis, D.L. Youngs, R.J.R. Williams, Growth of a Richtmyer–Meshkov turbulent layer after reshock, *Phys. Fluids* 23 (9) (2011) 095107.
- [33] J. Larsson, S.K. Lele, Direct numerical simulation of canonical shock/turbulence interaction, *Phys. Fluids* 21 (12) (2009) 126101. <https://doi.org/10.1063/1.3275856>
- [34] A.N. Kolmogorov, The local structure of turbulence in incompressible viscous fluid for very large reynolds numbers, *Proc. Math. Phys. Sci.* 434 (1890) (1991) 9–13. 51980
- [35] Y. Zhou, A scaling analysis of turbulent flows driven by Rayleigh–Taylor and Richtmyer–Meshkov instabilities, *Phys. Fluids* 13 (2) (2001) 538–543.
- [36] G. Dimonte, D.L. Youngs, A. Dimits, S. Weber, M. Marinak, S. Wunsch, C. Garasi, A. Robinson, M.J. Andrews, P. Ramaprabhu, A.C. Calder, B. Fryxell, J. Biello, L. Dursi, P. MacNeice, K. Olson, P. Ricker, R. Rosner, F. Timmes, H. Tufo, Y.-N. Young, M. Zingale, A comparative study of the turbulent Rayleigh–Taylor instability using high-resolution three-dimensional numerical simulations: the alpha-group collaboration, *Phys. Fluids* 16 (5) (2004) 1668–1693.
- [37] M. Groom, B. Thornber, Reynolds number dependence of turbulence induced by the Richtmyer–Meshkov instability using direct numerical simulations, *J. Fluid Mech.* 908 (2021).
- [38] D. Livescu, J.R. Ristorcelli, Buoyancy-driven variable-density turbulence, *J. Fluid Mech.* 591 (2007) 43–71. <https://www.cambridge.org/core/journals/journal-of-fluid-mechanics/article/buoyancydriven-variable-density-turbulence/12528EA6CABCA410E85F1CDBDAF7B9CE>. <https://doi.org/10.1017/S0022112007008270>